Occupant-Adaptive Airbag Deployment System (OAADS)

An Intelligent Multi-Stage Restraint Architecture Guided by The Toyota Way

A collaborative project of Presidency University and Toyota Kirloskar Motor
Project Team: Steve Johnson, Musharaf Baig, Dhananjaya Reddy, Sagar N

1. Executive Summary and Philosophy

Our starting point was not a laboratory bench but a roadside reality. Genchi Genbutsu, the Toyota Way practice of going to the actual place to see the actual situation for oneself, taught us that restraint systems are usually evaluated on a crash-test dummy that represents almost nobody. In practice, the same fixed-force deployment meets a 45 kg teenager, a 60 kg adult, and a sleeping child leaning toward the dashboard — and it answers all three with identical force. That is precisely how a safety device becomes a source of injury.

This project therefore sets out to do one thing well: give the airbag a sense of who it is protecting. The Occupant-Adaptive Airbag Deployment System (OAADS) is an intelligent, multi-stage restraint architecture that senses occupant characteristics in real time and adapts inflation force dynamically, so that the energy released matches the person receiving it. The work sits directly beneath the goal of Toyota Safety Vision 2030: Zero Fatalities — not as an add-on to that vision, but as one of the practical steps toward it.

2. Problem Statement

Jidoka is the principle that quality must be built into the process itself, rather than inspected into the product afterwards. Measured against that standard, conventional restraint systems fall short whenever they meet an occupant outside the narrow profile they were designed around:

- Fixed-force failure. The airbag is ejected at a constant, unyielding force that takes no account of the passenger's mass, stature, or seating position.
- Vulnerable occupants. Rigid deployment thresholds place children and lighter passengers at elevated risk of severe neck and chest trauma, because the cushion's energy is calibrated for a heavier body.
- A parked safety state. Conventional systems have no intelligent sleep/wake interlock tied directly to engine status and occupant presence, so the restraint logic is effectively blind between trips.

These three gaps describe a system that is built to a fixed assumption rather than to the people who actually sit in the seat.

3. Systems Engineering Framework

The project follows a rigorous systems engineering path that runs from the silicon block through to the vehicle ecosystem, organised across eight core pillars:

- **Purpose:** dynamic staged inflation, optimised throughout for injury mitigation rather than for a single pass/fail threshold.
- **Boundary:** scoped from the ESP32 microcontroller and its embedded edge processing out to the vehicle safety actuators.
- **Requirements:** strict functional safety and quality standards, treated as binding constraints rather than aspirations.
- **Decomposition:** a modular design that divides responsibility cleanly between simulation environments and physical prototyping.
- **Interfaces:** deterministic timing contracts built on I2C, GPIO, and sensor telemetry.
- **Partitioning:** physical placement distributed across the microcontroller, the edge inference unit, and the pneumatic hardware.
- **Failure Modes (FMEA):** degrade-and-recover protocols in which sensor loss safely triggers a robust baseline fallback mode.
- **Evolution:** trust-enabled over-the-air (OTA) calibration updates, so the system can improve after it leaves the factory.

4. Execution Workflow: The Four-Stage Autonomous Pipeline

At runtime the system moves through a deterministic, high-speed loop built to respect the critical time-to-fire (TTF) constraint. Each stage hands the next a settled answer rather than a guess:

1. **Sensing** — payload mass is collected in real time through load cells and piezoresistive arrays, while proximity and distance vectors are gathered by ultrasonic and radar sensors.
2. **Perception** — the system classifies occupant weight category, seating posture, and occupant count.
3. **Analysis** — the onboard processor weighs crash severity against posture data to calculate the precise force stage required.
4. **Deployment** — staged actuation opens proportional pneumatic valves, matching the cushion's energy to the occupant profile the system has just identified.

Classification is completed before the valve opens, so the deployment decision is already informed by posture rather than corrected after the fact.

5. Quantitative Validation and Simulation Results

Simulink studies and physical bench prototypes were used together to test whether the concept holds up under realistic conditions. The headline results are encouraging, and are summarized below.

Metric	Result	Reference point
Chest compression deflection	Reduced by 42%	Fixed-stage airbags, forward-leaning occupants
Posture classification accuracy	98.6%	12,000 diverse anthropometric posture variations
Decision pipeline latency	14 ms	Meets the sub-20 ms threshold required prior to TTF
Head Injury Criterion (HIC-15)	680 down to 395	Simulated 50 km/h frontal impacts

Taken together, the numbers describe a restraint that is gentler on the occupant without being slower to act — the latency budget is met with roughly 6 ms to spare, and the reduction in chest deflection and HIC-15 both point in the same direction.

6. Hardware Implementation and Prototyping

The prototype deliberately keeps the bill of materials modest, so that the intelligence of the system is demonstrated by the control logic rather than by expensive hardware:

- **Processing unit** — an ESP32 microcontroller managing core sensor polling and Blynk IoT telemetry.
- **Sensor suite** — an HX711 amplifier paired with a mass sensor, plus an HC-SR04 ultrasonic sensor for distance tracking.
- **Actuator control** — a PCA9685 16-channel PWM driver managing precision servo-controlled pneumatic valves alongside an air pump relay module.
- **Cloud visibility** — real-time monitoring of mass, distance, and ejection force percentages through the Blynk app dashboard.

7. Limitations and Continuous Improvement

Kaizen asks that we name our limits, because a limitation that is acknowledged can be worked on and one that is hidden cannot. Two areas are worth stating plainly.

Out-of-bound scenarios. Unconventional seating positions — such as an occupant resting in reverse against the deployment module, or leaning out of a window — can introduce calculation anomalies, because the sensor geometry no longer describes the occupant's actual relationship to the cushion.

Future scope. Three directions follow naturally from the present work:

- Impact-adaptive multi-stage ejection.
- AI-driven crash severity forecasting.
- Post-impact autonomous safety protocols

8. Respect for People and Mandatory Safety Protocols

Aligned with Global NCAP guidelines and Toyota Safety Vision 2030, this project rests on a simple conviction: technology exists to protect human life, and it works best when it reinforces — rather than replaces — fundamental safety habits.

- Seatbelts are mandatory. Adaptive airbags complement, but never replace, standard three-point seatbelts, which keep occupants inside the protected kinematic corridor.
- Clearance maintenance. Occupants must keep a safe operating distance from dashboard panels so that the cushion has room to expand properly.
- Eliminating out-of-position risks. Avoiding slouching and unauthorised seating postures prevents severe internal shear forces during dynamic events.

Project Team and Acknowledgements

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